

AI Critique

Overall, Claude performed effectively when generating API test cases for HW6. Across all three features, the AI-generated test suites successfully covered domain partitions, state transitions, security constraints, and schema validations. Most test cases were valid, featuring well-defined request payloads and deterministic expected results. The AI even succeeded in discovering some interesting edge cases for the API tests. However, because the number of test cases for each API endpoint was large, test case quality was not maintained consistently across all given test cases. The AI began to hallucinate functional requirements not documented in EShop's README or the API specification. For instance, in the registration suite, it wrongly assumed whitespace trimming requirements on parameters. In the cart suite, it incorrectly demanded error codes for lowercase authorization headers, despite headers being case-insensitive, and mistakenly included a test case (TC21) that was meant for FR08 instead of FR07. The incorrect feature target test case (TC21) was caused because the AI tried to come up with a new test case to cover state transitions, without noticing that it had targeted the wrong feature. Additionally, several test cases had non-deterministic expected outcomes, which is problematic for automated testing. After collaborating with the AI in this homework, I have learned a few lessons to mitigate these issues in future testing tasks. Firstly, I will refine my prompting strategy and explicitly instruct the AI to target the correct feature. Secondly, I will include notes about possible mistakes so that the AI can avoid these ahead of time.